

coding approaches are forward error correction (FEC) and automatic repeat request (ARQ).

Error control for a one-way system is accomplished using FEC. For two-way communication, for example, telephone channel and some satellite communication systems, ECC is accomplished using error detection and retransmission called ARQ. Generalised classifications of ARQ and FEC are shown in Figs 2 and 3, respectively.

Taxonomy of FEC techniques

Block codes. In this coding scheme, each k -bit information block is converted to an n -bit coded block with $(n-k)$ redundant bits added to k -bit symbols. These redundant bits are referred as parity bits that do not carry any information. The resulting code is referred to as (n, k) block code. Here, the code rate is defined as k/n .

In binary block codes, 2^k k -bit message sequences are uniquely mapped into 2^n n -bit codes, out of possible 2^n n -bit codes.

Convolutional codes. In this, a code-word depends not only on the corresponding k -bit message block at the same time, but also on m previous message blocks. Hence, the encoder has a memory order of m . Because the encoder contains memory, it is implemented with sequential logic circuits. Convolutional codes are widely accepted, and advancement or improvement to this coding scheme have resulted in two new coding schemes, namely, Trellis coded modulation (TCM) and turbo codes.

Coded modulation techniques. In both block and convolutional coding schemes, coding gain is achieved with the price paid for the bandwidth. Since in these schemes k -bit information signal is replaced by n -bit coded words ($n > k$), the required bandwidth gets increased, which is a major bottleneck for band-limited chan-



Fig. 4: Voyager 1, launched in 1977 by NASA (Image courtesy: <http://voyager.jpl.nasa.gov>)

nels such as telephone channels.

To overcome this problem, combined modulation and coding schemes are considered. In this case, coding gain is achieved with the price paid for the decoder complexity. Different coded modulation techniques are Trellis coded modulation, block coded modulation, multi-level coded modulation and turbo coded modulation.

Application areas

Deep Space communication. Without an error-correction scheme, Voyager and Galileo would not have been able to send the volume of data that they did, using less transmitting power. They used a concatenation of Reed-Solomon code and convolutional code.

Many early applications of coding were developed for National Aeronautics and Space Administration



Fig. 5: Various storage media

(NASA) and European Space Agency (ESA) deep Space and satellite communication systems where power was very expensive and bandwidth was plentiful.

A deep Space mission is expensive and additional cost for developing and implementing complex ECC encoder and decoder is required, which can be tolerated. Each decibel of coding gain realised resulted in an overall saving of about one million dollars (in 1960's) in transmitting and receiving equipment. Because of this, mathematician Massey